

ABHIRAM RACHAMADUGU

Mechanical Engineering — Robotics, Autonomy & Aerospace Systems
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EDUCATION

University of Illinois Urbana-Champaign — B.S. Mechanical Engineering Aug 2023 – May 2027 (expected)
Urbana-Champaign, IL • GPA 3.36 / 4.00 • Technical GPA 3.45 / 4.00

Mallya Aditi International School — AICE Diploma Jun 2019 – May 2023
Bengaluru, India • GPA 4.00 / 4.00

ENGINEERING EXPERIENCE

Technical Founder — Taiyo Aerospace Pvt Ltd • *Bengaluru, India* Jan 2025 – Present
- Founded an startup to engineer autonomous turbine-powered UAVs for civilian search-and-rescue and emergency response.
- Lead the end-to-end design and fabrication of a fixed-wing platform, managing the integration of micro turbojet propulsion systems.

Vice President & Fixed-Wing Lead — Illini Autonomous Vehicles • *Urbana-Champaign, IL* Jan 2024 – Present
- Founded UIUC's sole autonomous drone engineering team, directing all fixed-wing R&D initiatives.
- Designed and fabricated a custom carbon-fiber monocoque fuselage, maximizing the payload-to-weight ratio for a large-scale UAV.
- Developed a highly stable, tailless flying wing glider, successfully implementing self-stabilizing airfoils to optimize passive flight control.

Forward Deployed Engineer — Independent • *Bengaluru, India* May 2026 – Present
- Rebuilt aartiforgirls.org from a JavaScript-heavy build into a static Jekyll site, with a content overhaul.
- Built a document sorter for LEED and GRIHA certification: Python extraction, LLM classification, Excel output over 10,000+ files.
- Built a water-balance calculator solving networks of hundreds of flows from a draw.io flowchart.

New Product Initiatives Intern — Aiti Motors Pvt Ltd • *Bengaluru, India* Sep 2024 – Apr 2025
- Led development of a new light-duty autonomous mobile robot (AMR) for education and warehousing customers.
- Wrote the autonomy codebase in ROS 2 and deployed it across three units.
- Deployed and maintained two AMRs in the field, covering the full product lifecycle through after-sales support.
- Trained a successor and documented technical workflows before handing the project over.

Engineering Intern — ABB India • *Bengaluru, India* May 2024 – Jul 2024
- Designed several automotive assembly lines.
- Programmed robot routines for spot-welding car bodies.
- Maintained a large array of industrial robots on the factory floor.

Research Associate — VeraTatva Engineering Consultants LLP • *Bengaluru, India* Aug 2021 – Mar 2023
- Designed a low-cost solar-biomass hybrid vegetable dryer using CFD and thermodynamic simulation.
- Deployed 30+ units to marginal farmers in Kadapa, India with the NGO Aarti for Girls.
- Opened a sales channel for those farmers through a partnership with Amano Foods India, a sauce and soup manufacturer.

PERSONAL PROJECTS

Low-Cost 3D Printed Bionic Arm • *Bengaluru, India* Apr 2021 – Jan 2022
- Built several iterations of a 3D printed bionic arm at a unit cost under \$200.
- Used print-in-place (PIP) design for a fast-printing, material-efficient gauntlet requiring no assembly.
- Trained a compact machine learning model for voice control and exposed it through an API on the arm.

Pneumatic Exoskeleton • *Kadapa, India* Jul 2020
- Built an exoskeleton from metal tubing and pneumatic cylinders capable of lifting 80 kg (180 lb).
- Controlled actuation with an Arduino driving solenoid valves.

LEADERSHIP & COMMUNITY WORK

Initiative Lead — Children for Children • *Kadapa, India* Aug 2020 – Present
- Raised \$120,000 for COVID-19 relief in Kadapa, India, and distributed food packets to 1,000 families during the second wave.
- Launched a separate scholarship programme in 2021, awarding \$150,000 to date to 1,090 children orphaned by the pandemic.
- Continue to run the scholarship programme, managing fundraising, selection and disbursement.

TECHNICAL SKILLS

Programming Python (Advanced), C (Advanced), Java (Intermediate), C# (Intermediate), SQL (Intermediate)

Robotics & Autonomy ROS 2, ABB RobotStudio, Arduino, embedded control

CAD Fusion 360 (Advanced), SolidWorks (Advanced), CATIA (Advanced), Rhinoceros (Intermediate)

Simulation CFD, thermodynamic modelling, DesignBuilder (Advanced), Grasshopper (Intermediate)

Fabrication Metalwork (Advanced), thermoplastics & FDM (Advanced), woodwork (Advanced), carbon fiber layup, 5-axis CNC

PUBLICATIONS

- Article in the World Sustainable Energy Days (WSED) conference journal.
- Article in *Shashwat*, published by GRIHA, India's building certification board.